Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

Project: LingoLive — Understand. Reply. Keep going.

Team name: CreativeEngine

Team leader: Ashish T. Vasant | Member: Janhvi Singh

Problem: Help immigrants and interstate newcomers respond in an unfamiliar language.

Theme: Media, Content & Digital Experiences | Category: Education

Brief about the idea

Find your words in a new city.

A newcomer in Bengaluru needs to answer a cab driver in Kannada. LingoLive helps them understand, choose a reply and say it aloud.

Understand

Live multilingual speech → meaning

Reply

Useful options + pronunciation

Learn later

Saved words and personal practice

Opportunities

How is it different?

Connects the next turn, a pre-task plan, relevant personal/place memory, and later learning in one small interface.

How does it solve the problem?

Prioritizes usable replies during the encounter; the learner reviews word meanings afterwards. Repeated replies do not become new questions.

USP and intended users

Local, editable context follows the learner. Initial pilots: relocation teams, universities and community language groups. Outcomes and demand are not yet measured.

List of features offered by the solution

Hear and respond
Google multilingual transcription; three short replies.

Say it your way
Express familiar-language intent in the learning language.

Recognize readbacks
Fuzzy repeat guards + session voice estimates; correction.

Plan before leaving
Cab → optional tea → station; explicit progress controls.

Relevant memory
Editable routines; opt-in saved-place notes on return.

Learn afterwards
Saved review, word bubbles and pronunciation.

Minimal interface
One large microphone control; typing and first-use tutorial.

Bounded access
Google sign-in, no passcode; shared daily limits.

Process flow diagram or Use-case diagram

1. Listen

16 kHz browser microphone

→

2. Transcribe

Gemini Transcribe Live

→

3. Route the turn

Text/voice repeat estimates

Your repeated words

Keep in local conversation context. Wait for the other speaker. Uncertain meaning → confirm the speaker.

Other / unknown turn

Two Flash-Lite branches → meaning + replies → learner speaks → Saved learning later.

Speaker estimates help routing; they are not reliable identity. Explicit correction always wins.

Wireframes/Mock diagrams of the proposed solution (optional)

Conversation

Heard words

Meaning in your language

Three usable replies

Original / Pronunciation

Large listen button

Before the task

Where do you want to go?

Optional stops

Learner-controlled progress

Relevant local memories

Saved learning

Your conversation library

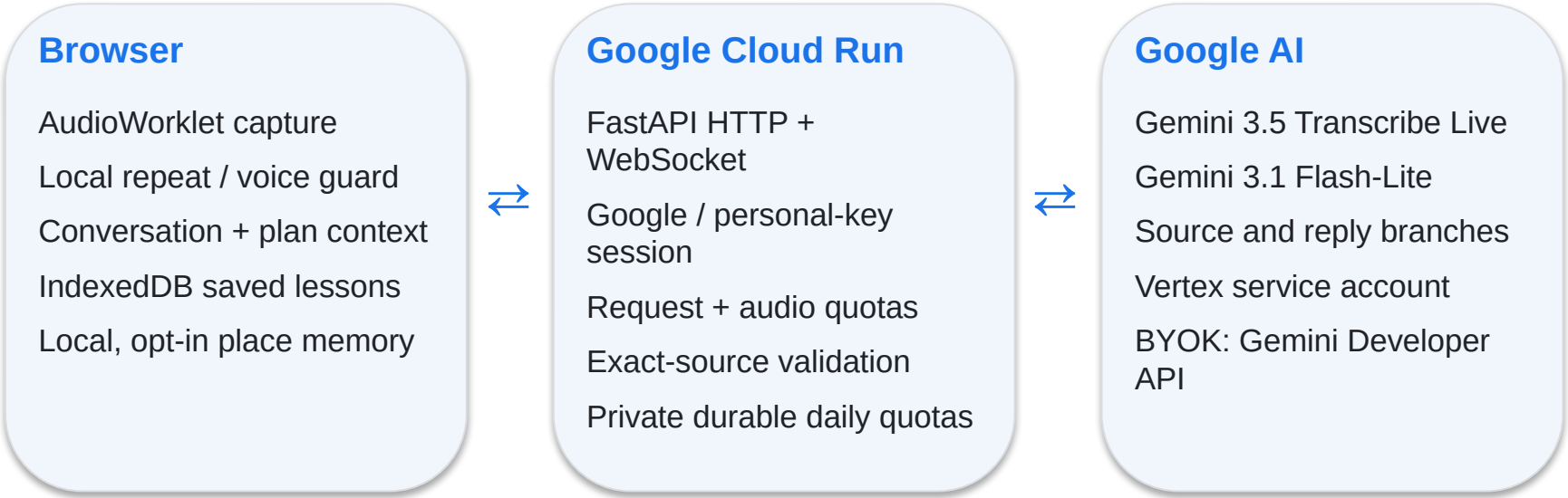
Tap a word → meaning

Pronunciation and practice

No new lesson request

Functional prototype screens are shown in the snapshots section and video.

Architecture diagram of the proposed solution



Firebase Authentication: Google sign-in. Private Cloud Storage: daily quotas. Firestore is not used; learning stays local.

Technologies to be used in the solution

Google AI

Gemini 3.5 Transcribe Live; two parallel Gemini 3.1 Flash-Lite branches with minimal thinking.

Google Cloud

Firebase Authentication, Cloud Run, Vertex AI, private Cloud Storage and Secret Manager.

Web and backend

Responsive JavaScript PWA, AudioWorklet, IndexedDB, Python FastAPI and WebSockets.

Local adaptation

Bounded context and lexical/acoustic heuristics; no model-weight fine-tuning or stored voiceprint.

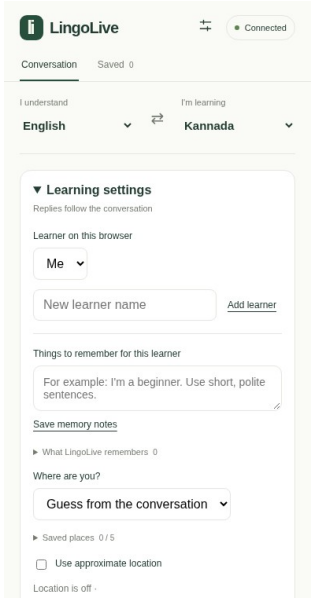
Estimated implementation cost (optional)

Cost model: pay for useful turns, bound prototype usage.

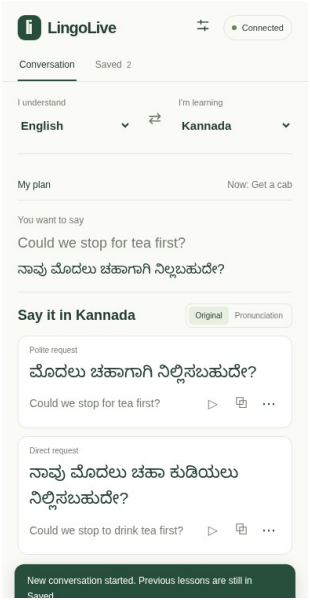
- A 15-minute listening session uses up to 15 minutes of streaming audio.
- N new lessons use 2 × N Flash-Lite generation calls.
- Detected repeats, saved word review and script switching add no lesson call.
- Cloud Run: 1 CPU / 512 MiB, request billing, min 0 / max 1 instance.
- Shared/day: 20 listening minutes + 100 lessons. Reset: midnight IST.
- Prices depend on tokens, provider rates and actual runtime; no dollar estimate is asserted.

Persistent quotas survive restarts; they do not cap all hosting/storage/download charges in currency.

Snapshots of the prototype



Plan before leaving



Find a useful reply



Learn afterwards

Recorded prototype views. Word-review screenshot uses labelled test data; the video includes real Google responses.

Prototype Performance report/Benchmarking

Functional evidence, with clear measurement limits.

Earlier v1.5 warm HTTP lessons	10 complete lessons · mean 1.79 s · historical sample
Later v1.9 typed HTTP checks	4 functional cases · mean 2.065 s · not a latency guarantee
Current readback guard	25 local checks + 10 browser-flow checks; repeats add no lesson
Voice estimates	3/4 held-out clean synthetic learner clips recognized; 0/4 others labelled user
Service checks	134 backend tests; Google-token verification and durable quota races checked

No measured learning-outcome, noisy-phone, cold-start or population speaker-accuracy claim.

Additional Details/Future Development (if any)

Social value and business case

Help newcomers participate in everyday encounters.

Pilot with relocation teams, universities and community language groups.

Proposed organization-sponsored access; validate demand before expansion.

Next engineering work

Representative mobile/noisy-language studies.

Opt-in local neural voice enrollment and durable per-user budgets.

Opt-in cross-device memory, potentially Firestore.

Measure task completion, reply usefulness and learning retention.

Current limits: ASR can mishear; voice estimates can be wrong; local profiles are not private cloud accounts.

Provide links to your:

GitHub source — github.com/U4AR/LingoLive

Demo video — 3–4 minute working demonstration

Final product — live Google Cloud Run prototype

Submission materials — pitch, technical PDF and entry text

Keep the source repository publicly accessible throughout evaluation.

AI access: sign in with any verified Google account; no passcode. Your own Google key is optional.

Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.

Thank you

